

Organisms as Entropy-Production Agents

Revising Schrödinger's Negentropy Thesis: Life as Nature's Most Efficient Instrument of Entropy Increase

I. Introduction

Between 2009 and 2010, Arto Annala and his collaborators published a series of groundbreaking works that systematically established the thermodynamic foundations of evolutionary theory. In "Physical Foundations of Evolutionary Theory" (2010), Annala and Salthe explicitly demonstrated that Darwin's theory of natural selection can be subsumed by the Second Law of Thermodynamics — evolution is not a special process independent of physical laws, but an inevitable manifestation of energy dissipation and dispersal in open systems. By redefining probability as a physical quantity that includes both combinatorial configurations and conditional circumstances, Annala extended statistical physics from conserved systems to open systems, bringing biological processes and physicochemical processes under a unified thermodynamic framework.

Building upon this paradigm, Annala further articulated a profound insight: **there is no essential difference in thermodynamic principles between living and non-living systems — both are mechanisms of energy dissipation, with the only distinction being that living systems incorporate history through information storage (DNA).** Organisms assemble mechanisms through genetic information to access and consume free energy rapidly, yet fundamentally, biological processes are thermodynamic processes.

In 2025, inspired by Annala's framework, independent researcher Zou Zhikai published a work describing the universe as a discrete energy network. This work concretized Annala's energy-gradient-driven principle into a computable discrete model, and derived a corollary of profound biological significance: **The essence of life is a spontaneously evolved system for maximizing cumulative entropy production through path planning.**

This view does not deviate from Annala's framework but rather specifies and operationalizes its core insight — if Annala proved that "life is entropy-producing," Zou demonstrates "how life becomes a more efficient entropy-producing agent by storing and planning high-cumulative-entropy paths."

II. The Core of Annala's Framework:

Energy-Gradient-Driven Evolution

The central contribution of Annala and Salthe (2010) is the formulation of the entropy principle as an equation of motion, whose meaning is remarkably simple and profound: **energy flows from heights to lows as soon as possible**. Systems evolve along the steepest gradients of free energy until reaching the free energy minimum in their respective surroundings — the thermodynamic stationary state.

Annala explicitly identified several key characteristics of this process:

First, evolutionary paths are inherently unsolvable. Because open systems have no conserved quantities, driving forces and flows are interdependent — flows alter the energy landscape, and the landscape in turn redirects flows. Thus, evolutionary processes are non-Hamiltonian, and their trajectories cannot be analytically determined in advance. This is not a lack of knowledge but a fundamental non-determinacy of the system.

Second, natural selection is the selection of energy flows. The largest energy flows naturally select the channels that level off energy density differences in the shortest time along the steepest directional gradients. Mechanisms that achieve the fastest energy dissipation are "selected" by the energy flows themselves and prevail in competition.

Third, life enjoys no privilege. Thermodynamics draws no demarcation line between animate and inanimate. Living systems are merely more sophisticated dissipative structures that incorporate history through information storage (DNA), enabling efficient access to and consumption of free energy — yet in principle, they are no different from energy dissipation processes in inorganic systems.

Fourth, all life phenomena are expressions of entropy production. Expansion, proliferation, differentiation, diversification, and catalysis — these are all ways for a system to evolve toward the stationary state in its respective surroundings. Competition and cooperation are not ends in themselves; whichever mechanism best contributes to entropy production is naturally selected by the flows of energy.

III. From Local to Cumulative Entropy Production: Zou's

Development

Annala's equation describes the entropy production rate at each moment — the system always selects the steepest gradient direction under current conditions. However, Zou introduces a critical distinction:

Local instantaneous entropy production refers to a system's entropy yield in the very next moment. This is a capacity shared by all physical systems — living or non-living.

Cumulative long-term entropy production refers to total entropy generated over multiple time steps. This is not a capacity shared by all systems, as it requires **foresight, path-planning ability, and mechanisms for storing experiential templates**.

Zou's discrete network model further reveals that at any given moment, a system may face multiple path choices with **nearly equivalent local entropy production rates** — yet over extended time horizons, the cumulative entropy they generate can diverge dramatically.

This provides the fundamental rationale for the existence of life: **organisms capable of identifying and selecting high-cumulative-entropy paths hold a decisive competitive advantage in evolution**.

IV. Why Cumulative Entropy Confers Evolutionary

Advantage

Annala has already noted that proliferation increases a system's mechanistic capacity to diminish free energy, catalysis accelerates the transduction process, and adaptation improves a system's consumption of available free energy. Zou further unfolds this logic, revealing the specific pathway through which cumulative entropy production operates in Darwinian competition:

Energy is the hard currency of survival. All biological activities require free energy consumption. The rate and total amount of energy an organism acquires, transforms, and dissipates directly determine its biomass, reproductive rate, and competitive capacity.

High cumulative entropy paths mean higher energy throughput. Choosing behavioral paths that generate greater cumulative entropy over time enables an organism to process more free energy across its lifespan. This high energy throughput translates directly into higher metabolic rates, faster growth, and greater reproductive output.

The pathway to evolutionary advantage is clear: organisms select high cumulative entropy paths → higher energy intake and dissipation → larger size or earlier

maturation → more offspring with higher survival rates → gene frequency rises, establishing evolutionary advantage.

Natural selection is essentially a filter among cumulative entropy paths. Annala characterized natural selection as a "competitive exclusion principle" — once a mechanism gains a slight advantage in energy dissipation, positive feedback accelerates it further, eventually excluding other competing paths. This principle applies not only among molecular species but also between individuals and economic rivals. Zou articulates this insight as: organisms that more efficiently identify, remember, and execute high-cumulative-entropy paths gain overwhelming advantages. Those locked into low-cumulative-entropy paths are replaced by more efficient entropy agents.

V. What Differentiates Organisms: Divergent Cumulative Entropy Rates

Annala and Salthe (2010) observed that skewed distributions are ubiquitous in nature — whether species abundance, genome sequence lengths, or protein lengths, their cumulative curves are sigmoid, appearing as power laws on log-log plots. This indicates that the same thermodynamic principle — maximum entropy partition — produces consistent statistical characteristics across different scales and systems.

Zou applies this insight to understanding differences among organism types: **the fundamental difference between species lies not in whether they produce entropy, but in the rate at which they do so, along which paths, and over what time scales.**

Viruses exhibit extremely low cumulative entropy rates, entirely dependent on host molecular machinery with single, constrained paths.

Bacteria, through rapid reproduction and local diffusion, achieve notable energy dissipation, yet remain limited by path complexity.

Insects, through simple approach-avoidance behaviors, achieve moderate entropy rates.

Mammals, with complex learning and memory, flexibly adjust behavioral strategies to sustain high cumulative entropy rates in changing environments.

Humans push this capacity to its extreme — through cultural and technological accumulation, externalizing high-entropy paths into tools, machines, and knowledge systems, achieving exponentially increasing cumulative entropy rates.

At every level, the "competitive exclusion principle" described by Annala is at work: **species capable of higher cumulative entropy rates gradually replace those with lower rates** — not because entropy has selective preferences, but because higher entropy rates mean higher energy throughput, higher reproduction, and greater environmental shaping capacity.

VI. Life Strategies as Expressions of Entropy Paths

Annala and Salthe (2010) identified multiple ways systems dissipate free energy: expansion extends the spatial scope of dissipation, proliferation increases the number of dissipative mechanisms, differentiation opens new dissipation channels, catalysis accelerates dissipation, and adaptation maintains or improves dissipation efficiency in the face of competitors. Zou's framework further explains how organisms optimize these entropy paths through specific strategies:

Memory and learning essentially store previously successful high cumulative entropy paths. Annala noted that biological information (DNA) enables living systems to incorporate history and rapidly assemble dissipative mechanisms. Zou extends this insight to the neural level: a proven foraging route or empirically validated predator-avoidance strategy is a validated high-entropy-path template. Organisms preserve these templates through genetic encoding and neural plasticity, enabling future direct invocation without re-exploration.

Cooperation and altruism expand the dimensionality of entropy paths. Annala observed that competing systems turn to collaboration when their integration yields an increase in entropy production. Zou specifies how this thermodynamic principle manifests in biological societies: social animals integrate individual local entropy behaviors into group-level efficient dissipation networks through cooperation. A wolf pack hunting cooperatively achieves a cumulative entropy rate far exceeding the sum of individuals hunting alone. Altruism, in thermodynamic terms, represents individual sacrifice of local entropy for enhanced group cumulative entropy — which ultimately benefits individual gene propagation.

Technology and culture make humans the only species to externalize and permanently store high-entropy paths. Annala noted that standards — such as the genetic code and chiral consensus — emerge to facilitate energy dispersal: pure-handed systems require only one reaction type to advance energy transduction, while mixed-handed systems require multiple catalysts. Similarly, international standards, technical norms, languages, and currencies are "standards" that facilitate larger-scale energy dissipation. Zou extends this to: the use of fire, the invention of agriculture, the mastery of electricity, the establishment of information technology — each technological revolution represents a dimensional upgrade in humanity's cumulative entropy paths, exponentially increasing the efficiency of this entropy-producing agent.

VII. The Thermodynamic Logic of Entropy Agents

Winning in Evolution

Integrating Annila's thermodynamic-evolutionary unification with Zou's cumulative-entropy-path perspective, we can formulate a unified thermodynamic account of evolutionary advantage:

Reproduction is entropy production expressed at the biological level. Annila noted that genotypes that reproduce fastest in given conditions come to prevail in the gene pool, as fast reproduction means faster free energy dissipation. Zou further articulates that each reproductive event is a large-scale energy reconfiguration — parental energy is redistributed to offspring, creating new energy distribution patterns. Higher reproductive rates mean higher cumulative entropy rates for a species within its niche.

Natural selection is competition among different entropy paths. Annila formulated this as the "competitive exclusion principle": once a mechanism gains a slight advantage in energy dissipation, positive feedback accelerates it further, eventually excluding other competing paths. Zou expresses this from a path-selection perspective: those paths generating higher cumulative entropy in a given environment enable their carriers (organisms and their genes) to acquire more resources and produce more replicators, thereby occupying larger population proportions.

Evolutionary advantage essentially equals the ratio of a species' cumulative entropy-production rate, within a given ecological environment and time window, to that of competing species. A ratio greater than 1 means expansion; less than 1 means decline. The entire history of evolution is, as Annila describes, a sustained rate competition among different entropy-producing agents in a thermodynamic process.

VIII. Conclusion

The work of Annila and Salthe (2010) established the thermodynamic foundations of evolutionary theory, revealing that natural selection is an expression of the Second Law of Thermodynamics, and that living and non-living systems share no essential difference in physical principles. They demonstrated that: **energy gradients drive evolution, maximum entropy production paths are naturally selected, and the diversity of life is the expression of diverse energy dissipation pathways.**

Building upon this framework, Zou (2025) introduced the concept of "cumulative entropy path planning," discretizing and rendering computable Annila's continuous thermodynamic framework, and further articulating: **the fundamental advantage of organisms lies in their capacity to store historical high-entropy paths and make forward-looking selections among multiple locally equivalent paths to achieve greater cumulative entropy production.**

The two perspectives are highly complementary. Annila answers "why life is entropy-producing" — because all systems are entropy-producing, and life is no

exception. Zou answers "how life becomes a more efficient entropy-producing agent" — through storing, planning, and invoking high-cumulative-entropy paths, organisms surpass the entropy-production efficiency of pure physical processes.

As Annala quotes Boltzmann: "The animate struggle for entropy, not against it." From the rapid reproduction of bacteria to the global expansion of human civilization, the biological world presents a sustained race of cumulative entropy-production rates. And the final winner is always the entropy-producing agent with the greatest foresight and the most refined capacity to store and invoke high-cumulative-entropy paths.

References:

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